

Forestry Sciences

Gopal Shukla · Abha Manohar K. ·
Anil Raj Kizha · Pankaj Panwar ·
Sumit Chakravarty *Editors*

Forest Degradation and Management

An Indian Perspective

 Springer

Forestry Sciences

Volume 87

Series Editor

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and Science and Genetics
Castilla La Mancha University
Albacete, Spain

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ISSN 0924-5480

Forestry Sciences

ISBN 978-3-031-84054-8

<https://doi.org/10.1007/978-3-031-84055-5>

ISSN 1875-1334 (electronic)

ISBN 978-3-031-84055-5 (eBook)

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Preface

The forests of India, which range from the great Himalayas to the Western Ghats, are not only a means of sustenance for millions but also a legacy. About 21.71% of the country's geographical area is under forests, which are at a very critical juncture today. The increasing threats of forest depletion and destruction constitute new challenges to environmental security, biodiversity maintenance, and achieving sustainable development in the country.

This volume, entitled *Forest Degradation and Management: An Indian Perspective*, is particularly timely as the urge to comprehend, evaluate, and counter forest degradation has become extremely important. It aims to advocate the integration of various aspects of forestry science and practice to represent a more comprehensive view of forest degradation processes and their impacts emerging from Indian forests.

This compilation, which aims at establishing the preliminary context of the Indian forests, begins with a comprehensive description of the causes and processes of deforestation and forest degradation and their analysis in space and time. This in turn opens the way for examining the more specific causes of these changes that are taking place in the forested areas, which are both man-made and natural. The contribution of climate change and the role of carbon emissions to local forest degradation provide an essential understanding of the global impacts of local processes. What is most appealing about this volume is the presence of different perspectives. The comparison of these classical and modern evaluation techniques is beneficial to both practitioners and researchers, as it traces the growth of forest monitoring processes. Such technical precision is complemented by a concern for ecology, exploring how such degradation may propagate at the ecosystem level as functional loss, biodiversity erosion, and the breakdown of vital ecosystem services. Managing Indian forests requires addressing soil health issues as well as forest degradation. A focused study of herpetofauna in the Western Ghats biodiversity hotspot makes a compelling argument about how forest degradation quickly impacts specific taxonomic groups. The forest regeneration impacts reveal the impacts of forest over-stress on the outlook of our forest ecosystems.

A significant strength of this volume is its emphasis on the integration of scientific analysis and management aspects. The research findings presented in each chapter are based on field research and include practical aspects that may be useful to forest managers, policymakers, and conservationists. In this manner, the reading of the book is constructive as it contributes to both theory and practice of forest management in India. This construction comes at a time when India is focusing on its commitment to steps that would strengthen the cause of conservation and restoration of forests, as indicated in our national objectives of increasing and enhancing forest cover and decreasing forest degradation. It is the intention of the authors to write a book that would be necessary in implementing these aims using logic and managerial approaches.

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Abha Manohar K. is a dedicated forestry professional with a strong passion for research, education, and forest conservation. Her expertise lies in the areas of forest regeneration, medicinal plant cultivation, and sustainable forest resource management. She holds a Ph.D. in Forestry from Uttar Banga Krishi Viswavidyalaya, West Bengal, India, where her research focused on developing propagation protocols for endangered medicinal plants. She also graduated from Kerala Agricultural University and obtained her postgraduate degree from SHUATS, Uttar Pradesh. She brings a wealth of experience to her work, having held positions as both a researcher and an Assistant Professor. In these roles, she has developed valuable skills in conducting research, managing projects, and teaching forestry courses. Her research has been published in numerous peer-reviewed journals and book chapters, and she is a recognized expert in the field of forest regeneration. She is a strong advocate for sustainable forest management practices. She is particularly interested in finding solutions to the problem of forest degradation, and her work focuses on developing methods for restoring degraded forests and promoting the sustainable use of forest resources. She is also passionate about supporting rural livelihoods and believes that sustainable forestry practices can play a key role in improving the lives of forest-dependent communities. In addition to her academic and research pursuits, she is actively engaged with professional organizations and serves as a reviewer for several scientific journals. She has also delivered invited lectures at various seminars, conferences, and training programs and has chaired technical sessions at conferences.

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Part I

Forest Degradation: Assessment

Trends of Deforestation and Forest Degradation in India and Its Impact



Abhilash Dutta Roy, Sumit Kumar Das, Gopal Shukla, Pankaj Panwar, and Sumit Chakravarty

Abstract India, in 2023, surpassed China to become the most populous country. The geographical area of India is comparatively lower, and it is the seventh-largest country in terms of total geographical area (TGA). However, it supports more than 17% of the global population. Due to the extreme population pressure, large-scale urbanization, growing infrastructure, and rapidly growing economy, the pressure on existing forests is increasing. Prompt corrective measures have been taken to ensure the safekeeping of existing forest resources, and its positive effects have been observed in many parts of the subcontinent. However, challenges still exist, especially in the northeastern parts of the country where, due to cultural factors and population pressure, the area under forest cover is still declining at an alarming rate. Considering all these factors, the present chapter tries to shed light upon the past trends, the current status, and future directives in the context of the two significant challenges, i.e., deforestation and forest degradation.

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1 Introduction

The earth is experiencing a rapid and concerning depletion of forest cover, primarily attributable to anthropogenic pressure. From 2000 to 2012, a staggering 2 million km² of forests were cleared globally, with a predominant portion of this loss observed in tropical regions. For the prevention of further deforestation, various measures have been instituted on both local and global fronts. These include initiatives such as Payment for Ecosystem Services (PES), Reducing Emissions from Deforestation and Forest Degradation (REDD+), and international agreements to curb illegal timber trade. Furthermore, areas characterized by dense vegetation and ecological sensitivity have been restricted as protected zones. In developing nations, promoting community-based forest management has emerged as a strategy to ensure the sustainable preservation of these vital ecosystems (Ranjan 2018). In the Indian context, biodiversity conservation is intricately entwined with population well-being, given that a considerable number of forest fringe and Indigenous communities reside proximate to or within designated conservation areas, relying mainly on the ecosystem services provided by the forests. The substantial population density and commensurate biomass needs inherent to these households pose a substantive risk of inducing forest degradation and consequent biodiversity diminution (Davidar et al. 2010).

Forest resources constitute fundamental components within ecosystems, offering not only tangible provisioning services in the form of timber and non-timber forest products (NTFPs) but also an array of regulating and supporting services, including watershed protection, nutrient cycling, pollution mitigation, microclimatic regulation, and carbon sequestration. They also provide cultural services, including spiritual and religious values, to the local communities through sacred groves. The degradation of these invaluable resources exerts localized and global repercussions on the economy and the environment. Indeed, in developing nations like India, forest degradation impedes future growth and development. The tangible costs of such degradation manifest through its impact on food security and water availability. While macro-level impacts may not be readily apparent, the issue assumes critical significance at the regional and local levels. At the local and microlevel, social and institutional factors mediate the impact of broader structural variables like anthropogenic pressures and market forces on forest loss (Agrawal 1995). It implies that the effectiveness of local institutions is more influential in determining resource conditions than population pressure or market forces alone. Without effective forest governance to curtail degradation, the realization of United Nations (UN) sustainable development goals (UN SDGs) remains elusive, as sustaining natural capital (Kn) is imperative for long-term viability (Panwar et al. 2022). The depletion of forest resources warrants scrutiny from both quantitative and qualitative perspectives (Reddy et al. 2001).

India has been grappling with the complex interplay of human activities and their impact on forest cover. Over the past decades, the country has substantially transformed its forested areas, primarily influenced by population pressure, urbanization,

agricultural expansion, and other anthropogenic disturbances (Chakravarty et al. 2012). The negative correlation between economic growth and forest cover in India is evident (Shahani and Khaneja 2021). The growth in economic activity, coupled with the increasing population, has led to increased demand for land, resulting in the loss of crucial forested areas. In the quest to achieve economic growth targets, India faces the daunting task of reconciling its commitment to the National Forest Policy of 1952, which aimed to cover 33% of the country's geographical area under forest. The current trajectory suggests a deviation from this goal, emphasizing the urgency of policymakers to adopt strategies that harmonize economic development with forest conservation. Our overview highlights the challenges that India is facing to maintain its forest cover and biodiversity while pursuing economic development. The urgency for sustainable forest management practices, community engagement, and innovative conservation strategies is paramount to address this delicate balance and secure the future of India's diverse ecosystems. Even under significant market and population pressures, it has been seen in India that smallholders can demonstrate the capacity to conserve forest resources. It is, however, contingent upon a supportive sociopolitical context that encourages autonomous local action and incorporates effective institutional incentives for sustainable environmental use (Agrawal 1995). Macro-level correlations may not be sufficient for explaining changes in resource conditions. It is necessary to understand causal processes at the microlevel to formulate effective policy measures for addressing deforestation and forest degradation in India.

2 Recent History of Deforestation in India

The forests of the Indian subcontinent have been through severe anthropogenic disturbance recently. The colonial history of deforestation and forest degradation in India spans from the nineteenth century, marked by the establishment and expansion of railways during the British Raj. The development of Indian railways, often considered a cornerstone of resource extraction for the British colonial state, played a pivotal role in shaping the ecological landscape. Railway construction aimed to facilitate efficient resource extraction from India to support the British industrial capitalist economy. The demand for timber, particularly for railway sleepers, led to widespread deforestation, notably in the Himalayan region, with profound consequences on India's natural resources (Das 2011). India faced deforestation challenges in the early twentieth century, with approximately 14 million hectares lost due to agricultural expansion, timber extraction, and increasing energy demands (Chakravarty et al. 2012; Singh et al. 2014).

The colonial state established the Forest Department in 1864 across India. This intervention primarily was aimed to ensure a cost-effective timber supply for the railways. However, at the time, the Forest Department could not provide a sustainable, long-term solution to the ecological challenges posed by colonial resource extraction policies (Das 2011). Historians are divided on the impact of colonial

forestry policies, particularly those associated with creating the Indian Forest Service (IFS). While some argue that the IFS had a conservationist ethic focused on climatic and proto-ecological considerations (Barton and Bennett 2008), others contend that the IFS was inherently linked toward a capitalist and imperialist approach, contributing to large-scale deforestation (Joshi 1983; Guha and Gadgil 1989).

Introducing technologies such as dams and railways exacerbated environmental problems, contributing to the qualitative break in forest denudation from the pre-colonial to the colonial phase (Barton and Bennett 2008). The ecological consequences of British colonial rule, including deforestation, were not uniform across regions and evolved, reflecting the complex interplay of economic interests and environmental considerations (Das 2011). However, it has to be noted that conservation in the form of protected areas was also introduced during this time. The Jim Corbett National Park, covering an area of 325 km², came into being as India's first and the world's third National Park in 1936. Colonial exploitation left a lasting impact, and the initial years after independence saw continued deforestation. India's post-independence economic growth from the 1950s to the 1980s, dominated by agriculture, contributed to significant deforestation (Singh 2011). The country's forest cover declined by 12.4 million hectares between 1950 and 1980. Forest exploitation for timber, agricultural expansion, and infrastructure development were primary drivers. Forest policies, including large-scale felling and diversion for non-forestry purposes, added to the challenges. The transition began in the 1980s, marked by the stabilization of the forest estate and subsequent recovery. India's economic shift toward the service sector, rural-urban migration, and reduced pressure on land resources played pivotal roles (Singh et al. 2014). The Forest Conservation Act of 1980 curtailed the diversion of forest lands, initiating a turning point. Social forestry programs, afforestation efforts, and international support contributed to stabilizing and recovering India's forests (ICFRE 2010). The National Forest Policy of 1988 was also adopted during this time. In the twenty-first century, India is witnessing a gradual decrease in deforestation rates. Our results in Sect. 4 show that the loss of forest area between 2000 and 2022 is around 3.5 times less than the forest area lost between 1950 and 1980. Planted forests and afforestation programs have been crucial in achieving this result. However, challenges persist in the form of unregulated fuelwood consumption, illicit timber harvesting, unregulated livestock grazing, and fodder collection, contributing to ongoing forest degradation (Davidar et al. 2010). Studies highlight the impact of localized degradation, reflecting the economic importance of the forest sector in peri-urban settings (DeFries and Pandey 2010). Policies like the Forest Rights Act (2005) have aimed to address forest dependency and promote sustainable management, but anomalies remained. The forest landscape evolved, with primary and natural forests remaining comparatively stable in most parts of the country except in the northeastern region.

3 Forest Cover Change in India Estimated Through Remote Sensing

Over the past decades, anthropogenic pressure has negatively affected forest cover in India. Human land use constitutes 83% of land area change in India, as opposed to the global 55% (Meiyappan et al. 2017). Anthropogenic disturbances such as agricultural expansion, logging, urbanization, infrastructural development, and other land use are the major factors contributing to the loss of vegetation and forest cover (Kumari et al. 2020). The forest cover in India has been changing rapidly due to increasing human activity and overexploitation, particularly in high population density areas (Moyra and Harza 2016). Remote sensing (RS) technology is a pivotal tool in assessing and monitoring land use and land cover (LULC) change, including forest cover dynamics. It offers us valuable insights into the impact of human activities on Indian forests at a fast and reasonably accurate forest cover data over large geographical areas.

Satellite data and socioeconomic databases have been used to examine the dynamics of forest cover change in India. The country experienced a net decline in forest cover by about 18,000 km² between 1985 and 2005 (Meiyappan et al. 2017). The use of multispectral and synthetic aperture radar (SAR) remote sensing has allowed for the estimation of forest cover and greenness in India, with results indicating that significant loss of forest cover has occurred in India between 2007 and 2017 (33,291.59 km² or 5.85% loss), while the increase in green cover is often due to plantations rather than the expansion of forest covers (Lal et al. 2019). However, it is worthy of note that over the same period, the Forest Survey of India has recorded a net gain of 17,384 km² or 2.51% gain (FSI 2007, 2017). This disagreement in forest cover change indicates a potential knowledge gap that may be explored.

Species composition has undergone a notable transformation in the continuously diminishing forested landscapes. The composition of forests in the Himalayan foothills exhibited an increase in the area of commercially important plantation species such as Chir pine (*Pinus roxburghii*) and Khair (*Acacia catechu*), while a decline in the area of native oak (*Quercus leucotrichophora*) and bamboo (*Dendrocalamus strictus*), which are facing high anthropogenic pressure, has been seen (Shah and Sharma 2015). Long-term remote sensing of forests coupled with available historical maps has also provided valuable information on the distribution and changes in forest cover, as well as the rate of deforestation in India. Forest cover suffered a net loss of 243,447 km² (28%) in 8 decades from 1930 to 2013 (Reddy et al. 2016). Predicted forest cover maps of 1880–2025 show that 416,443 km² or 40% of the forest cover in India was lost during the time interval of 1880–2013, with the highest forest cover loss in Maharashtra, followed by Odisha, Karnataka, and Assam (Reddy et al. 2017).

The impact of economic growth on forest cover suggests a negative relationship between forest area and gross domestic product (GDP) in India, indicating that a decrease in forest cover is associated with a rise in economic growth (Shahani and Khaneja 2021). Furthermore, the growth trend in forest cover from 1987 onward

indicates that India is unlikely to fulfill its target of covering 33% of the country's geographical area under forest cover, as outlined in the National Forest Policy of 1952 (Chaudhry and Tambe 2023). It portrays a concerning picture to address this issue, and policymakers must adopt strategies that balance economic growth with forest conservation. Sustainable forest management practices and the promotion of agroforestry, forest certification, and financial incentives such as more significant involvement in the carbon exchange market for facilitating payment for ecosystem services (PES) can contribute to maintaining and enhancing forest cover while fostering economic development. Public awareness and participation are also vital components in ensuring the sustainable use of forest resources.

4 Recent Trends in Forest Cover Change (2000–2022): An Analysis of the Hansen Global Forest Change data

For analyzing the net loss of forest cover in India, Global Forest Change (v1.10) (GFC) raster data developed by Hansen et al. (2013) has been used. The Global Forest Change (GFC) data were developed using Landsat 30 m spatial resolution data on a global scale, which undergoes annual updates, making it a reliable source of forest change on a worldwide scale. We have used Google Earth Engine (GEE) JavaScript API to calculate the recent trends in forest cover change from 2000 to 2022 using the 'loss,' 'gain,' and 'loss year' bands. The 'loss' band shows the forest cover loss for each year, while the 'gain' band shows the forest gain for that year. Our analysis considered areas with at least 10% canopy cover as forests, including natural and planted forests. We calculated the overall loss for each year using the 'loss' band and the net loss using both the 'loss' and 'gain' bands. Preliminary analysis showed that forest loss was much more significant than gain, so we subtracted the gain band from the loss band for each year to calculate the net annual loss. Gain and loss trends for the entire country along with individual states and union territories within India were worked out.

The total overall forest loss from 2000 to 2022 in India was 38,297.27 km² or 3.83 million hectares, while the total net forest loss was 36,641.35 km² or 3.66 million hectares. The highest net annual loss occurred in 2016 and 2017, with 3644 and 3918 km² of forest cover lost, respectively. There is a statistically significant trend ($R^2 = 0.73$) of increase in annual forest loss with time (Fig. 1). Contrastingly, forest gains were minuscule at 1655.92 km², with high gains observed only in the earlier years. Over the same period, the population of India grew by 34%, from 1.06 billion in 2000 to 1.42 billion in 2022 (World Bank 2023). This demographic growth accentuates the intricate challenges of harmonizing forest conservation with the dynamic demographics of the country, emphasizing the pressing need for incentives to follow sustainable forest management to address this delicate balance.

Among the states and union territories, the highest net loss was recorded in Mizoram, with 6341.71 km²; Assam, with 5956.33 km²; and Nagaland, with

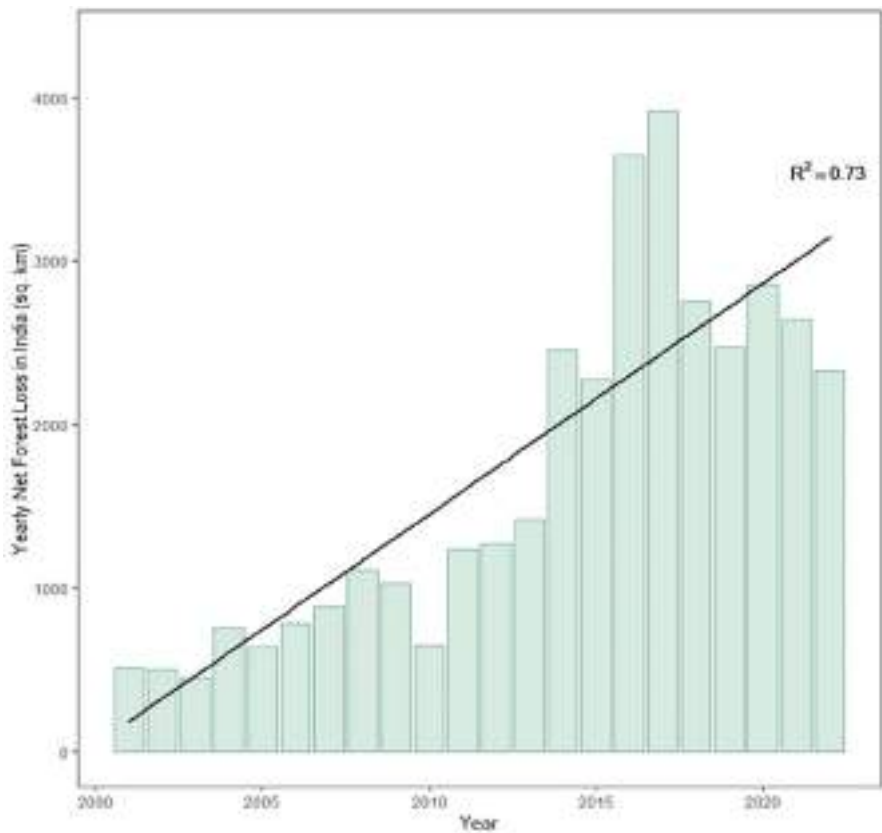


Fig. 1 The yearly net forest loss in India (2000–2022) shows a statistically significant upward trend over the years

5633.92 km² (Fig. 2). It is interesting to note that the states with the highest forest loss also have high forest cover percentages within state areas. Most of the states with the highest net forest loss are in Northeast India, which is renowned for its rich biodiversity and fragile ecosystems.

The repercussions of diminishing forest cover in Northeast India extend beyond the loss of trees; they directly threaten the species that call these forests home, including the local Indigenous communities. The delicate balance of ecosystems is disrupted, affecting local flora and fauna and, in turn, impacting the livelihoods of local communities dependent on these ecosystems for sustenance. The increasing cultivation of plantation crops such as oil palm, rubber, and areca nut further exacerbates the challenges, signaling a transformation in the traditional way of life for these communities.

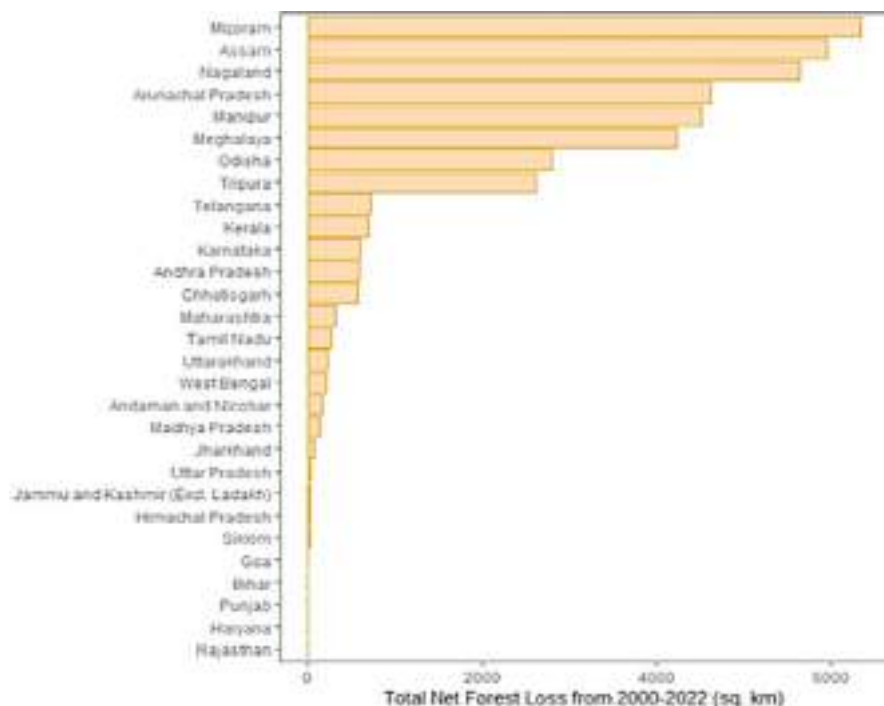


Fig. 2 The state-wise net forest loss in India between 2000 and 2022

5 Impact of Anthropogenic Disturbance on Biodiversity in Indian Forests

Deforestation constitutes a primary catalyst for the extinction of biodiversity in tropical regions. The ramifications of deforestation within tropical biodiversity hotspots are particularly alarming due to the concentration of globally endemic species in these areas. Despite this, there is a dearth of comprehensive documentation on the effects of extensive deforestation on native biotas in the Himalayan biodiversity hotspot. India is a treasure chest of biodiversity that hosts many plants and has been identified as one of the eight primary centers of origin and crop diversity. It accounts for 7–8% of the global biodiversity, with an estimated 5150 plant species and 1837 animal species as endemic (Venkataraman 2012). Being the center of origin of many domesticated species, India's biodiversity includes wild relatives of many crops and domesticated animals. The country has 16 major types and 251 subtypes of forests. As one of the 17 identified megadiverse countries, India has 10 biogeographical zones and is home to 9% of the known mammalian species.

However, India has seen a tremendous deforestation rate due to population pressure in the past centuries and has four biodiversity hotspots: the Himalayas, Indo-Burma, Sundaland, and Western Ghats. However, this trend is being seen in other

countries, too. Globally, the prospects for tropical forests and the biodiversity therein are becoming increasingly bleak, owing to unabated deforestation and forest alteration stemming from human activities such as logging, hunting, agricultural expansion, and human settlement (Giam 2017). It was projected that at the current deforestation rate, total forest cover in the Indian Himalayas will be reduced by the end of this century to about half of the forest cover as reported in 1970, and the dense forests will be reduced to only 34%. According to the extinction theory, larger and more specialized species are likely to be lost due to deforestation (Sodhi et al. 2004). A troubling trend of deforestation has been reported in the Indian Himalayas and forecasts the potential extinctions of endemic taxa (species and subspecies) by the year 2100 across diverse taxonomic groups, encompassing gymnosperms, angiosperms, fishes, amphibians, reptiles, birds, and mammals. Considering the current magnitude of deforestation, it is projected that by 2100, merely 10% of the Indian Himalayas' land area will be covered by dense forest (>40% canopy cover). Under such circumstances, almost a quarter of the endemic species face the risk of extinction, including 366 endemic vascular plant taxa and 35 endemic vertebrate taxa. Additionally, relevant findings underscore the significance of accurate reporting of forest cover data by governmental institutions. Inaccuracies in reporting lead to underestimations of the biological consequences of deforestation and have the potential to result in miscalculations in land-use decisions, such as the construction of hydroelectric dams (Pandit et al. 2007). Research indicates that species dependent on forest ecosystems typically struggle to adapt to habitat loss (Brooks et al. 2002). Consequently, it is imperative to conserve habitats and prioritize the management of species deemed vulnerable to these changes. Urgent and extensive conservation initiatives are imperative to mitigate the imminent biodiversity losses driven by deforestation.

6 Impact of Overgrazing in the Forests of the Indian Himalayas

Grazing in forests has been shown to significantly impact plant community structure and diversity. Overgrazing and under-grazing in montane forests have been seen to negatively impact vegetation community composition and are in accordance with the intermediate disturbance hypothesis (Kikoti and Mligo 2015). Forest degradation by overgrazing livestock can significantly impact the ecosystem, particularly in loss of pasture biomass and reduction in ecosystem carbon stocks (Eldridge and Delgado-Baquerizo 2017).

In addition to plant community structure and biomass, grazing impacts soil properties and hydrology. Livestock overgrazing has been seen to negatively impact soil properties in montane forests, where livestock grazing decreased the accumulation of soil organic carbon (SOC) and consequently negatively affected the natural pattern of availability of nitrogen, phosphorus, and cation exchange capacity (Mligo

2015). Selective browsing by animals has also been seen to lead to soil erosion, depressed growth of seedlings and saplings, and reduced recruitment due to trampling (Sharma et al. 2009). Overgrazing and browsing in forests can lead to the overconsumption of forage, which can harm the biodiversity and sustainability of the forest ecosystem. However, properly managed, targeted livestock grazing can have benefits, such as increased habitat diversity and regeneration of selected canopy tree species. However, these effects are context-dependent and require further research (Öllerer et al. 2019). Cattle grazing have positively impacted vegetation structure and dynamics, including germination, seed dispersal, flower and fruit production, and conservation of soil features (Mazzini et al. 2018).

India's cattle density is around 62 heads per km² compared to the global average of 10 heads per km² (Robinson et al. 2014). In the Himalayas, uncultivated fodders are an integral component of farming systems in the hills, with grass-dominated rangelands showing a good fodder value (Singh and Bohra 2005; Singh et al. 2008). The region's forests contribute significantly to the livestock sector in rural households and have implications for policy planning for poverty alleviation (Pandey 2011). However, rapidly growing livestock and human populations and unrestricted overgrazing have led to a decline in the density of vegetative cover in India. The increasing numbers of livestock can be attributed to the fact that the Indian Green Revolution had little impact on livestock production, and livestock pressure is a significant cause of forest degradation (Holden and Sankhayan 1998). In the Central Himalayas, the human population has been doubling every 27–30 years (Samal et al. 2003). As a result, it has been seen that the current forest fodder extraction rate is likely to be unsustainable (Pandey et al. 2014). Heavy grazing has reduced species diversity, increased the occurrence of ruderal and weedy species, and inhibited the regeneration of tree species (Kala and Rawat 1999; Thakur et al. 2011). The regular collection of fodder from the forests has been an essential factor in causing soil erosion, low fertility of the land, and other degradation processes in the hill areas (Chandra et al. 2008). Grazing has been seen to have altered the forest cover, affecting soil moisture regimes in the Sikkim Himalayas (Prokop and Płoskonka 2014).

The unavailability of fodder from other sources in the Himalayas may be the reason for the degradation of forest areas because these are more frequently used and are vital for the existing farming systems (Pandey et al. 2014). The strategy for reducing forest degradation must revolve around providing alternative fodder sources to the community for livestock rearing while encouraging the protection of the forest cover. Agroforestry can be a potential strategy to enhance climate change resilience among smallholders who are entirely dependent on natural resources for fodder in mountainous regions (Nagar et al. 2022). Silvopastoral systems have been recognized for their high nature value and biodiversity (Plieninger et al. 2021). Grazing in silvopastoral systems is essential to maintain the open savanna structure, but it can also help control shrub encroachment and maintain the natural regeneration of multiage stands (Bugalho et al. 2011). Policies that aid the local population to have easier participation in the global voluntary carbon market and certification

for sustainable forest management can help strengthen the local economy and thereby help reduce dependence on fodder for livestock.

7 Impact of Anthropogenic Disturbance on Mangrove Forests of India

Mangrove forests provide many ecosystem services and functions, including but not limited to timber and various NTFPs such as tannins, wax, honey, and aquaculture. They also boast vibrant soil organic carbon (SOC) pools sequestered up to ten times more blue carbon than above-ground pools. Mangroves are rich-blue carbon pools, with a global above- and below-ground carbon stock of mangrove forests of 1.6 Peta grams (Pg) and 10.2 Pg, respectively (Kauffman et al. 2020). These plants also play a significant role in climate change adaptation by stabilizing coastal lands through their complex structure and in areas that permit landward accretion of organic buildup, significantly reducing the impact of sea level rise. The dependence of coastal people on mangrove forests for direct use due to the scarcity of alternate resources may cause these fragile ecosystems to become highly disturbed. These include resource extraction, such as fuelwood consumption (Ambastha et al. 2010), and the expansion of shrimp monoculture (Datta and Deb 2017; Kumar 2012). The impact of these anthropogenic activities on land use change further exacerbates the loss of mangroves (Kumar 2012).

India lost 428 km² or 8.4% of its mangrove forests between 1980 and 2009 (Dasgupta and Shaw 2013). From 1975 to 2005, approximately 75 km² of Indian mangroves have been converted for aquaculture, and 171 km² for agriculture (Giri et al. 2008). For example, in the Bhitarkanika mangrove of Odisha, there was a significant extraction of resources such as fuelwood despite the forest being a protected area (Ambastha et al. 2010). In the Indian Sundarbans of West Bengal, the composition of significant mangrove species such as *Heritiera fomes* and *Excoecaria agallocha* have witnessed a decline of about 10% from 1977 to 2015, mainly due to anthropogenic factors such as overexploitation of timber (Ghosh et al. 2016). The mangrove forests in the Indian Sundarbans have suffered a lot owing to forests being converted for agriculture (Dasgupta and Shaw 2013). In the mangroves of Maharashtra, urbanization and industrialization have been identified as the main drivers of anthropogenic impacts (Mugade and Sapkale 2014).

To ensure the resilience of mangrove forests against anthropogenic pressure, collaborative efforts between government bodies, forest fringe local communities, and environmental organizations are necessary. Community engagement and strict enforcement of regulations are crucial components of any effective solution. Mangroves managed by nongovernmental organizations collaborating with local forest dependents have previously shown good performance (Datta 2018). Participation of forest-dependent communities in forest resource management is also beneficial for the local economy (Bardhan 2021). In the face of escalating